Shanilka Haturusinghe

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EDUCATION

Eötvös Loránd University (ELTE), Hungary | *MSc in Data Science* **2026 September – Present**

- Stipendium Hungaricum Scholar with a full scholarship.

University of Kelaniya, Sri Lanka | *BSc (Hons) in Computer Science, First Class* **2020 – 2024**

- GPA: 4.0/4.0 *(Ranked 1st in the Faculty of Computing and Technology)*
- Specialization: Artificial Intelligence
- Awarded two gold medals in 2025 for Academic Excellence and Outstanding Performance by the Alumni Association of the University of Kelaniya and the Faculty of Computing and Technology, respectively.

IELTS | *English Proficiency* **August 2025**

- Overall band score: 8.5 (Listening 9.0, Reading 8.5, Writing 7.5, Speaking 8.0)

LEADERSHIP & ACTIVITIES

ELTE Data Science Club **Budapest, Hungary**
Project Leader *- Autonomous Vision & Drone Tracking Project 2026 September – Present*

PUBLICATIONS

- **Shanilka Haturusinghe**, Tharindu Cyril Weerasooriya, Christopher M. Homan, Marcos Zampieri, and Sidath Ravindra Liyanage. 2025. *Subasa – Adapting Language Models for Low-resourced Offensive Language Detection in Sinhala.* In *Proceedings of NAACL 2025: Student Research Workshop*, pages 260–270, Albuquerque, USA. ACL. DOI: 10.18653/v1/2025.naacl-srw.26
- U. P. Melder, **S. Haturusinghe**, and S. P. K. Arachchi. 2026. *Customer Segmentation and Churn Prediction in the Maritime Shipping Industry Using Machine Learning Techniques: A Sri Lankan Case Study.* In *2026 6th International Conference on Advanced Research in Computing (ICARC)*, Belihuloya, Sri Lanka, pages 1–6. DOI: 10.1109/ICARC68737.2026.11453542

TEACHING EXPERIENCE

University of Kelaniya **Colombo, Sri Lanka**
Lecturer (Temporary) *August 2025 – June 2026*

- Lecturer-in-Charge for `CSCI 31032 - Theory of Programming Languages`: delivered all lectures, office hours, assignments, grading, and examinations.
- Contributing lecturer for `CSCI 31022 - Machine Learning and Pattern Recognition` and `CSCI 11532 - Structured Programming I`.

Teaching Assistant & Demonstrator *February 2025 – July 2025*

- Served as TA for *DSCI 32012 - Advanced Database Applications* (70 students), *CTEC 22043 - Object Oriented Programming* (50 students), *AINT 32012 - Digital Image Processing and Computer Vision* (10 students), and *CSCI 22082 - Object-Oriented Programming II* (50 students).
- Conducted office hours, led lab sessions, and developed and graded assignments across all modules.

INDUSTRY EXPERIENCE

Embla Software Innovation (Pvt) Ltd **Colombo, Sri Lanka**
Software Engineer Intern *October 2023 – April 2024*

- Developed and implemented the "Scrapbook" feature for the Embla FT genealogy application with customizable page templates, editable text/photo elements, and comprehensive image and content management capabilities.
- Built full-stack components using C#/.NET Core backend and Angular 9/TypeScript frontend, enhancing existing architecture to support new functionality and user interactions.
- Collaborated with product teams using Agile methodologies, Jira, and Bitbucket; utilized Visual Studio, FastReport Designer, and Figma for development and design.

Xempler Pte. Ltd **Remote**
Intern – Software Developer *September 2022 – May 2023*

- Designed and implemented an image processing pipeline for Sri Lanka License Plate Detection using Python, TensorFlow, and YOLO models; applied OpenCV preprocessing techniques and integrated Tesseract OCR for alphanumeric text extraction.
- Designed and implemented a real-time chat feature for the IFAWPCA mobile app using React Native, integrating Firebase Realtime Database and Firebase Authentication for instant message synchronization and user management.

ACHIEVEMENTS & AWARDS

Gold Medal for Academic Excellence – Alumni Association of the University of Kelaniya 2025

Gold Medal for Outstanding Performance – Faculty of Computing and Technology, University of Kelaniya 2025

Google Cloud Research Credits (Award GCP19980904) – $1,000 in research credits 2024

Huawei Seeds for the Future – Selected as one of four top university students from Sri Lanka to visit China; received mentorship from industry experts on 5G, AI, Cloud Computing, and Digital Power 2023

PROJECTS

PersonaCraft.AI – Personalized AI Writing Assistant | *github.com/haturusinghe/persona-craft-ai*

- Developed an end-to-end system for creating personalized AI writing assistants through automated data collection, processing, and model fine-tuning with multi-platform web scraping capabilities.
- Built automated data extraction pipelines using Selenium for LinkedIn, Medium, GitHub, and custom web sources, with advanced text processing using chunking and embedding generation via Sentence Transformers.

- Engineered training pipeline supporting both Supervised Fine-Tuning (SFT) and Direct Preference Optimization (DPO) using TRL and Unsloth; implemented MLOps architecture with ZenML, MongoDB, Qdrant, and AWS SageMaker.
- Key Technologies: Python, ZenML, LangChain, GPT-4, Llama 3.1, TRL, Unsloth, MongoDB, Qdrant, AWS SageMaker, Docker

Real-Time Fashion Recommendation System | *github.com/haturusinghe/realtime-fashion-recommender*

- Developed an MLOps-driven fashion recommendation system using H&M's Kaggle dataset, implementing a 4-stage decoupled pipeline architecture for feature processing, training, and inference.
- Implemented a two-tower neural network for customer and item embedding generation; used CatBoost for ranking model training and Hopsworks Feature Store for embedding storage and retrieval.
- Integrated Hopsworks Model Registry and Model Serving for deployment, with an interactive Streamlit frontend for real-time customer recommendations and feedback collection.
- Key Technologies: Python, TensorFlow, Hopsworks, Sentence-Transformers, CatBoost, Polars, Streamlit

Deep Learning-based Real-Time Facial Age and Gender Detection System

- Designed and trained CNN models (separable convolutions, batch normalization, LeakyReLU, dropout) on the UTKFace dataset with data augmentation, achieving robust age regression and binary gender classification.
- Built an end-to-end inference pipeline with TensorFlow Lite supporting CPU, GPU, and NNAPI acceleration for on-device real-time inference on Android.
- Key Technologies: Python, TensorFlow, TF-Lite, Flask, Kotlin, Android Studio, OpenCV

SKILLS & TECHNOLOGIES

Programming Python, C#, TypeScript, Kotlin

ML & Deep Learning PyTorch, TensorFlow, Keras, Scikit-learn

LLMs & NLP Hugging Face Transformers, LangChain, LangGraph, Llama, Mistral, LoRA/QLoRA, SFT, DPO, RAG

MLOps & Experiment Tracking ZenML, AWS SageMaker, Weights & Biases, Docker

Data & Databases MongoDB, Qdrant, Pandas, Polars, PostgreSQL

Web & APIs FastAPI, Flask, Streamlit

Dev Tools Git, GitHub, Bitbucket, Jira

REFERENCES

Prof. S. R. Liyanage Professor, Department of Software Engineering, Faculty of Computing and Technology, University of Kelaniya
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